AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	11.2 / 0.696386 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-15 23:00 UTC
Location	29.5831, -95.0155
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	6 / 400	61.34 to 100; mean 88.1796	87.69 at 2026-06-15 22:55Z; 5 min before event
asos	Air temperature (temperature)	degC	6 / 400	23.8889 to 32.7778; mean 26.9196	27 at 2026-06-15 22:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	6 / 400	0 to 360; mean 117.675	0 at 2026-06-15 22:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	6 / 400	0 to 11.3178; mean 2.6931	0 at 2026-06-15 22:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	6 / 72	5851.89 to 5926.29; mean 5891.94	5888.41 at 2026-06-16 00:00Z; 1 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	6 / 72	32.0322 to 1546.05; mean 348.138	343.959 at 2026-06-16 00:00Z; 1 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	6 / 72	53.4497 to 69.5435; mean 60.4965	61.4199 at 2026-06-16 00:00Z; 1 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	6 / 72	1005.18 to 1014.43; mean 1010.42	1010.49 at 2026-06-16 00:00Z; 1 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	6 / 72	16.785 to 19.498; mean 18.3749	18.8984 at 2026-06-16 00:00Z; 1 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	6 / 72	-3.35346 to 1.58078; mean -0.756	0.180553 at 2026-06-16 00:00Z; 1 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	6 / 72	-1.42555 to 6.14875; mean 2.3095	0.0344543 at 2026-06-16 00:00Z; 1 h after event
openaq	Ozone (ozone)	ppm	15 / 978	-0.001 to 0.031; mean 0.0136	0.012 at 2026-06-15 23:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 212	5 to 110; mean 25.4906	14 at 2026-06-15 23:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	76 / 2662	-2.6 to 52; mean 7.1395	2.4 at 2026-06-15 23:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	3 / 214	3 to 100; mean 90.9907	100 at 2026-06-15 23:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	3 / 214	68 to 100; mean 90.1542	96 at 2026-06-15 23:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	3 / 214	0 to 22.21; mean 1.1974	1.77 at 2026-06-15 23:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	3 / 214	1005 to 1016; mean 1011.52	1011 at 2026-06-15 23:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	3 / 214	24.22 to 33.3; mean 26.9809	26.49 at 2026-06-15 23:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	3 / 214	0 to 358; mean 159.724	154 at 2026-06-15 23:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	3 / 214	0.07 to 5.67; mean 2.2242	0.98 at 2026-06-15 23:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	56 / 4088	0 to 46.2; mean 8.1436	4.7 at 2026-06-15 23:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	1 / 1	0.0259566 to 0.0259566; mean 0.026	0.0259566 at 2026-06-14 19:24Z; 27 h 35 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	1 / 1	0.000111617 to 0.000111617; mean 0.0001	0.000111617 at 2026-06-14 19:24Z; 27 h 35 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	2.03572e-05 to 2.03572e-05; mean 0	2.03572e-05 at 2026-06-14 19:24Z; 27 h 35 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	1 / 1	-6.10873e-06 to -6.10873e-06; mean -0	-6.10873e-06 at 2026-06-14 19:24Z; 27 h 35 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 3 h 50 min before event
tceq	Carbon monoxide (co)	ppm	3 / 213	0.1 to 1.1; mean 0.223	0.2 at 2026-06-15 23:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	11 / 716	-2.1 to 27; mean 5.3127	11.2 at 2026-06-15 23:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 219	-0.6 to 0.6; mean -0.1169	-0.1 at 2026-06-15 23:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-14 06Z 90.79, 12Z 80.9, 18Z 68.88, 06-15 00Z 84.58, 06Z 90.66, 12Z 91.53, 18Z 90.61, 06-16 00Z 92.41, 06Z 96.58, 12Z 91.69, 18Z 86.36, 06-17 00Z 89.38, 06Z 95.51
asos	Air temperature (temperature)	06-14 06Z 26.87, 12Z 28.73, 18Z 30.8, 06-15 00Z 28.29, 06Z 27.19, 12Z 26.23, 18Z 26.06, 06-16 00Z 26.16, 06Z 25.59, 12Z 25.68, 18Z 26.95, 06-17 00Z 26.31, 06Z 24.82
asos	Wind direction (wind_direction)	06-14 06Z 100, 12Z 139.1, 18Z 174.4, 06-15 00Z 136.3, 06Z 129, 12Z 162.7, 18Z 81.52, 06-16 00Z 66.47, 06Z 70.86, 12Z 158.2, 18Z 97.43, 06-17 00Z 102, 06Z 104.8
asos	Wind speed (wind_speed)	06-14 06Z 1.886, 12Z 3.792, 18Z 4.464, 06-15 00Z 2.847, 06Z 2.847, 12Z 2.915, 18Z 1.122, 06-16 00Z 1.634, 06Z 0.9995, 12Z 3.435, 18Z 1.999, 06-17 00Z 2.896, 06Z 3.708
noaa_gfs	500-hPa geopotential height (gh_500)	06-14 12Z 5912, 18Z 5926, 06-15 00Z 5922, 06Z 5922, 12Z 5904, 18Z 5892, 06-16 00Z 5887, 06Z 5891, 12Z 5872, 18Z 5865, 06-17 00Z 5854, 06Z 5858
noaa_gfs	Planetary boundary layer height (pbl_height)	06-14 12Z 279.8, 18Z 1388, 06-15 00Z 416, 06Z 204.8, 12Z 292.7, 18Z 401.6, 06-16 00Z 152.3, 06Z 63.45, 12Z 222, 18Z 394.4, 06-17 00Z 227.7, 06Z 135.3
noaa_gfs	Precipitable water (precipitable_water)	06-14 12Z 57.93, 18Z 59.18, 06-15 00Z 60.72, 06Z 59.78, 12Z 57, 18Z 64.57, 06-16 00Z 61.51, 06Z 63.19, 12Z 65.31, 18Z 63.25, 06-17 00Z 56.32, 06Z 57.19
noaa_gfs	Surface pressure (surface_pressure)	06-14 12Z 1013, 18Z 1013, 06-15 00Z 1011, 06Z 1013, 12Z 1012, 18Z 1012, 06-16 00Z 1010, 06Z 1010, 12Z 1009, 18Z 1009, 06-17 00Z 1006, 06Z 1006
noaa_gfs	850-hPa temperature (t_850)	06-14 12Z 18.53, 18Z 18.12, 06-15 00Z 19.17, 06Z 19.22, 12Z 18.74, 18Z 17.12, 06-16 00Z 18.75, 06Z 18.42, 12Z 17.75, 18Z 17.46, 06-17 00Z 19.16, 06Z 18.07
noaa_gfs	10-m eastward wind (u_10m)	06-14 12Z -1.246, 18Z -0.7478, 06-15 00Z -1.395, 06Z -0.3972, 12Z -0.3435, 18Z -1.222, 06-16 00Z -0.3578, 06Z -0.2767, 12Z 0.5291, 18Z 0.6634, 06-17 00Z -1.883, 06Z -2.395
noaa_gfs	10-m northward wind (v_10m)	06-14 12Z 2.473, 18Z 4.645, 06-15 00Z 3.325, 06Z 2.567, 12Z 2.108, 18Z 3.859, 06-16 00Z -0.6872, 06Z 1.405, 12Z 2.391, 18Z 3.794, 06-17 00Z 0.386, 06Z 1.448
openaq	Ozone (ozone)	06-14 06Z 0.01253, 12Z 0.01576, 18Z 0.02206, 06-15 00Z 0.01423, 06Z 0.01078, 12Z 0.009173, 18Z 0.01489, 06-16 00Z 0.007927, 06Z 0.003067, 12Z 0.01057, 18Z 0.01722, 06-17 00Z 0.01511, 06Z 0.02073
openaq	PM10 (pm10)	06-14 06Z 45.67, 12Z 33, 18Z 22.17, 06-15 00Z 33.06, 06Z 49.44, 12Z 41.22, 18Z 12.22, 06-16 00Z 20.89, 06Z 23.69, 12Z 18.33, 18Z 15, 06-17 00Z 18.89, 06Z 14.44
openaq	PM2.5 (pm25)	06-14 06Z 11.3, 12Z 7.508, 18Z 8.502, 06-15 00Z 9.755, 06Z 9.794, 12Z 7.832, 18Z 4.725, 06-16 00Z 8.224, 06Z 7.188, 12Z 4.172, 18Z 2.557, 06-17 00Z 2.774, 06Z 4.853
openweather	Cloud cover (cloud_cover)	06-14 06Z 7.667, 12Z 39.62, 18Z 78.39, 06-15 00Z 95.72, 06Z 99.88, 12Z 98.63, 18Z 99.89, 06-16 00Z 99.39, 06Z 99.94, 12Z 100, 18Z 100, 06-17 00Z 99.68, 06Z 88.6
openweather	Relative humidity (humidity)	06-14 06Z 89.67, 12Z 85.31, 18Z 73.33, 06-15 00Z 87.44, 06Z 91.29, 12Z 92.53, 18Z 94.11, 06-16 00Z 92.94, 06Z 94.89, 12Z 93.67, 18Z 90.71, 06-17 00Z 91.89, 06Z 93.73
openweather	Precipitation rate (precipitation)	06-14 06Z 0, 12Z 0.1675, 18Z 0.095, 06-15 00Z 0.02056, 06Z 0.02941, 12Z 3.015, 18Z 5.285, 06-16 00Z 0.9906, 06Z 1.258, 12Z 1.152, 18Z 0.6988, 06-17 00Z 0.6616, 06Z 0.86
openweather	Surface pressure (pressure)	06-14 06Z 1014, 12Z 1015, 18Z 1013, 06-15 00Z 1014, 06Z 1013, 12Z 1014, 18Z 1012, 06-16 00Z 1012, 06Z 1010, 12Z 1011, 18Z 1009, 06-17 00Z 1008, 06Z 1006
openweather	Air temperature (temperature)	06-14 06Z 27.25, 12Z 28.91, 18Z 31.77, 06-15 00Z 28.52, 06Z 27.42, 12Z 26.53, 18Z 25.47, 06-16 00Z 25.99, 06Z 25.49, 12Z 25.53, 18Z 26.65, 06-17 00Z 26.33, 06Z 25.09
openweather	Wind direction (wind_direction)	06-14 06Z 138, 12Z 182.1, 18Z 179, 06-15 00Z 163.3, 06Z 167.9, 12Z 185.7, 18Z 152.9, 06-16 00Z 120, 06Z 202.8, 12Z 184.9, 18Z 148.4, 06-17 00Z 118.6, 06Z 109.5
openweather	Wind speed (wind_speed)	06-14 06Z 2.003, 12Z 2.616, 18Z 3.604, 06-15 00Z 2.141, 06Z 2.489, 12Z 2.291, 18Z 2.137, 06-16 00Z 0.9033, 06Z 1.627, 12Z 1.811, 18Z 2.381, 06-17 00Z 2.419, 06Z 2.39

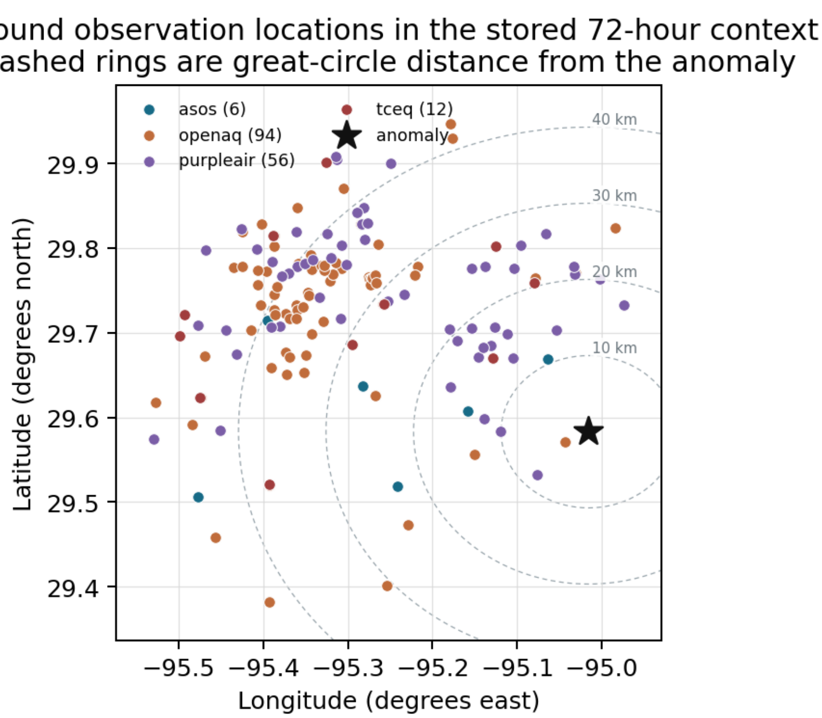
Source	Metric	Values
purpleair	PM2.5 (pm25)	06-14 06Z 12.4, 12Z 8.685, 18Z 10.73, 06-15 00Z 11.37, 06Z 10.67, 12Z 8.082, 18Z 6.695, 06-16 00Z 9.68, 06Z 9.278, 12Z 4.379, 18Z 5.161, 06-17 00Z 4.816, 06Z 7.477
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	CO granules available (s5p_co_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
tceq	Carbon monoxide (co)	06-14 06Z 0.2, 12Z 0.2111, 18Z 0.1933, 06-15 00Z 0.2167, 06Z 0.1778, 12Z 0.2333, 18Z 0.2667, 06-16 00Z 0.2778, 06Z 0.2278, 12Z 0.2556, 18Z 0.2444, 06-17 00Z 0.2167, 06Z 0.1556
tceq	Nitrogen dioxide (no2)	06-14 06Z 3.25, 12Z 2.18, 18Z 1.463, 06-15 00Z 2.342, 06Z 2.789, 12Z 5.323, 18Z 7.64, 06-16 00Z 10.77, 06Z 8.285, 12Z 5.369, 18Z 6.92, 06-17 00Z 6.024, 06Z 4.392
tceq	Sulfur dioxide (so2)	06-14 06Z -0.1667, 12Z -0.2333, 18Z -0.1278, 06-15 00Z -0.1833, 06Z -0.2389, 12Z -0.2111, 18Z -0.07222, 06-16 00Z 0.01667, 06Z -0.09444, 12Z -0.08333, 18Z -0.07222, 06-17 00Z 0.005556, 06Z -0.1

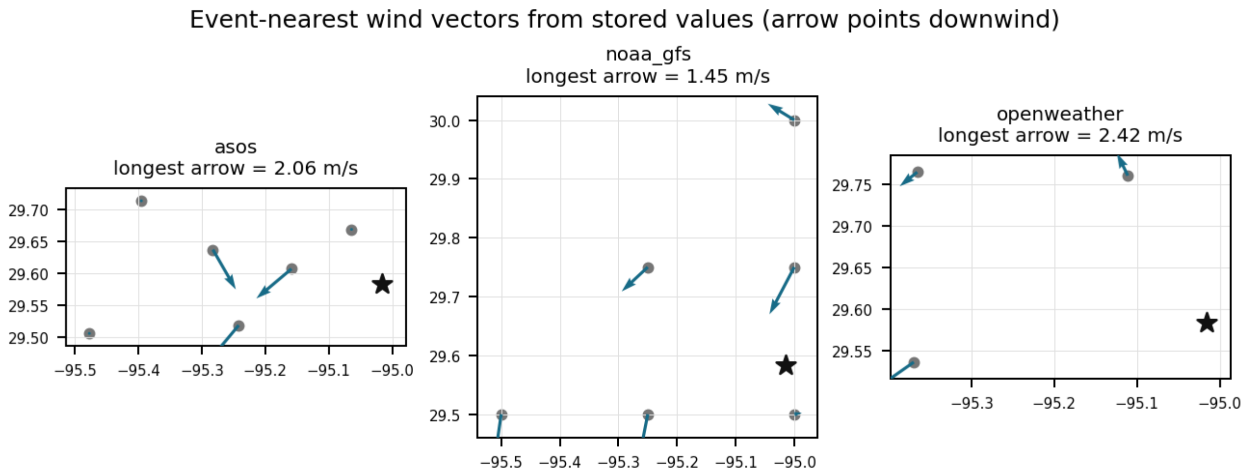
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	T41 (10.667 km) 87.44, EFD (14.102 km) 87.41, LVJ (23.01 km) 87.54, HOU (26.499 km) 87.63, MCJ (39.455 km) 92.52, AXH (45.443 km) 86.19
asos	Air temperature (temperature)	T41 (10.667 km) 27.25, EFD (14.102 km) 27.3, LVJ (23.01 km) 26.99, HOU (26.499 km) 27.15, MCJ (39.455 km) 26.38, AXH (45.443 km) 26.61
asos	Wind direction (wind_direction)	T41 (10.667 km) 148.5, EFD (14.102 km) 107, LVJ (23.01 km) 121, HOU (26.499 km) 137.5, MCJ (39.455 km) 126.1, AXH (45.443 km) 61.25
asos	Wind speed (wind_speed)	T41 (10.667 km) 3.315, EFD (14.102 km) 3.241, LVJ (23.01 km) 2.244, HOU (26.499 km) 2.965, MCJ (39.455 km) 3.601, AXH (45.443 km) 1.036
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.00 (9.357 km) 5892, gfs:29.75,-95.00 (18.624 km) 5891, gfs:29.50,-95.25 (24.49 km) 5892, gfs:29.75,-95.25 (29.288 km) 5891, gfs:30.00,-95.00 (46.387 km) 5891, gfs:29.50,-95.50 (47.768 km) 5893
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.00 (9.357 km) 436.5, gfs:29.75,-95.00 (18.624 km) 346.2, gfs:29.50,-95.25 (24.49 km) 341.9, gfs:29.75,-95.25 (29.288 km) 356.1, gfs:30.00,-95.00 (46.387 km) 294.2, gfs:29.50,-95.50 (47.768 km) 314
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.00 (9.357 km) 60.2, gfs:29.75,-95.00 (18.624 km) 60.72, gfs:29.50,-95.25 (24.49 km) 60.62, gfs:29.75,-95.25 (29.288 km) 60.77, gfs:30.00,-95.00 (46.387 km) 59.92, gfs:29.50,-95.50 (47.768 km) 60.74
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.00 (9.357 km) 1012, gfs:29.75,-95.00 (18.624 km) 1011, gfs:29.50,-95.25 (24.49 km) 1010, gfs:29.75,-95.25 (29.288 km) 1010, gfs:30.00,-95.00 (46.387 km) 1010, gfs:29.50,-95.50 (47.768 km) 1010
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.00 (9.357 km) 18.53, gfs:29.75,-95.00 (18.624 km) 18.37, gfs:29.50,-95.25 (24.49 km) 18.39, gfs:29.75,-95.25 (29.288 km) 18.32, gfs:30.00,-95.00 (46.387 km) 18.37, gfs:29.50,-95.50 (47.768 km) 18.27

Source	Metric	Values
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.00 (9.357 km) -0.6305, gfs:29.75,-95.00 (18.624 km) -0.5388, gfs:29.50,-95.25 (24.49 km) -1.042, gfs:29.75,-95.25 (29.288 km) -0.6355, gfs:30.00,-95.00 (46.387 km) -0.5521, gfs:29.50,-95.50 (47.768 km) -1.137
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.00 (9.357 km) 3.346, gfs:29.75,-95.00 (18.624 km) 2.102, gfs:29.50,-95.25 (24.49 km) 2.35, gfs:29.75,-95.25 (29.288 km) 1.789, gfs:30.00,-95.00 (46.387 km) 2.214, gfs:29.50,-95.50 (47.768 km) 2.056
openaq	Ozone (ozone)	5077791 (0.008 km) 0.01183, 332 (14.586 km) 0.01782, 2800 (21.051 km) 0.01432, 320 (24.786 km) 0.01212, 3214 (26.617 km) 0.01497, 339 (26.892 km) 0.01726, 2039 (28.563 km) 0.01144, 3378 (28.774 km) 0.009554, +7 more
openaq	PM10 (pm10)	271 (14.586 km) 20.73, 13380082 (28.774 km) 27.93, 15852419 (35.786 km) 27.75
openaq	PM2.5 (pm25)	5077638 (0.008 km) 10.67, 14404954 (2.969 km) 4.935, 13955815 (13.307 km) 6.365, 268 (14.586 km) 9.079, 2648 (20.93 km) 12.69, 11638043 (23.95 km) 3.503, 14157975 (24.789 km) 8.057, 2040 (28.563 km) 9.321, +68 more
openweather	Cloud cover (cloud_cover)	grid:east (21.774 km) 91.18, grid:south (34.665 km) 90.99, grid:center (39.395 km) 90.81
openweather	Relative humidity (humidity)	grid:east (21.774 km) 90.52, grid:south (34.665 km) 88.14, grid:center (39.395 km) 91.78
openweather	Precipitation rate (precipitation)	grid:east (21.774 km) 1.388, grid:south (34.665 km) 1.212, grid:center (39.395 km) 0.995
openweather	Surface pressure (pressure)	grid:east (21.774 km) 1012, grid:south (34.665 km) 1012, grid:center (39.395 km) 1011
openweather	Air temperature (temperature)	grid:east (21.774 km) 27.09, grid:south (34.665 km) 26.95, grid:center (39.395 km) 26.9
openweather	Wind direction (wind_direction)	grid:east (21.774 km) 166.5, grid:south (34.665 km) 155.1, grid:center (39.395 km) 157.6
openweather	Wind speed (wind_speed)	grid:east (21.774 km) 2.577, grid:south (34.665 km) 2.568, grid:center (39.395 km) 1.538
purpleair	PM2.5 (pm25)	2386 (8.161 km) 4.667, 3298 (10.11 km) 7.804, 247283 (12.095 km) 8.37, 268165 (12.925 km) 8.836, 222815 (13.892 km) 7.675, 222593 (15.865 km) 7.867, 128007 (15.888 km) 2.959, 161689 (15.943 km) 8.034, +48 more
tceq	Carbon monoxide (co)	482011039 (14.566 km) 0.1347, 482011035 (28.771 km) 0.1352, 482011052 (44.211 km) 0.4029
tceq	Nitrogen dioxide (no2)	482011050 (0.0 km) 1.451, 482011039 (14.566 km) 5.045, 482011015 (20.507 km) 4.806, 482010026 (26.633 km) 7.183, 482011035 (28.771 km) 8.671, 482010416 (29.325 km) 5.694, 480391004 (37.123 km) 2.29, 482011052 (44.211 km) 15.07, +3 more
tceq	Sulfur dioxide (so2)	482011039 (14.566 km) 0.02603, 482011035 (28.771 km) -0.4178, 482010051 (44.59 km) 0.0411




Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 2.06 m/s
noaa_gfs
longest arrow = 1.45 m/s
openweather
longest arrow = 2.42 m/s
29.70
29.65
29.60
29.55
29.50
−95.5
−95.4
−95.3
−95.2
−95.1
−95.0
30.0
29.9
29.8
29.7
29.6
29.5
29.75
29.70
29.65
29.60
29.55

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

Widespread cloud cover reaching 99 to 100 percent reduced photolysis rates, enhancing the localized accumulation of nitrogen dioxide emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 35

Meteorological conditions near 2026-06-15T23:00:00+00:00 favored near-surface NO_2 accumulation because the nearest ASOS station reported 0.0 m/s wind, openweather reported about 0.98 m/s wind, and gfs indicated a low boundary-layer height near 152 m by 00Z.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 35

A localized primary NOx emission source is more likely than a broad regional pollution event because PM2.5 remained low near the anomaly, ozone was modest, CO was not elevated, SO2 was not elevated, and no contemporaneous sentinel5p column evidence indicates a large regional plume over the area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 35

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 35

A severe thunderstorm or tornado event has stirred up dust and pollutants from the ground, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 35

Overcast cloud cover and high relative humidity diminished photochemical destruction of NO2, helping maintain elevated concentration levels near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 35

Localized combustion emissions from industrial facilities or transport sectors accumulated near the surface during weak atmospheric dispersion conditions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 35

The nitrogen dioxide reading of 11.2 ppb at 23:00 UTC on June 15, 2026, corresponds to a z-score of 17.5 above normal levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 35

The TCEQ monitor at 29.583, -95.016 recorded an NO2 concentration of 11.2 ppb at 2026-06-15T23:00:00+00:00, which was far above its expected value of 0.696 ppb and is consistent with a severe local anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 35

The PM2.5 levels were significantly higher than usual, with a mean value of 8.1436 ug/m3, indicating a high level of particulate matter in the air.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 35

The anomaly is located in the vicinity of Houston, Texas, with a distance of approximately 8.161 km from the nearest sensor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 35

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO_2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 35

Local accumulation of fresh NO_x emissions near the anomalous TCEQ monitor is a leading hypothesis because NO_2 rose sharply at 23:00 UTC while nearby winds were nearly calm, local wind speeds were below typical values, and boundary-layer mixing was shallowening into the evening.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 35

Short-range transport from nearby combustion sources such as roadway traffic, industrial equipment, marine or rail activity is a plausible cause because the NO_2 enhancement was strong at the anomalous monitor while regional particulate matter, carbon monoxide, and sulfur dioxide stayed low to moderate, which is consistent with a localized fresh NOx plume rather than a broad aged pollution episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 35

Stagnant surface winds and a collapsing planetary boundary layer severely restricted vertical and horizontal atmospheric mixing, allowing surface pollutants to accumulate.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 35

The air quality anomaly in Houston, Texas on June 15th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 35

Meteorological stagnation supports accumulation of locally emitted NOx because the nearest ASOS station reported 0.0 m/s wind at 2026-06-15 22:55 UTC, nearby openweather wind speed was about 0.98 m/s, and gfs boundary-layer height decreased from about 402 m at 2026-06-15 18Z to about 152 m at 2026-06-16 00Z under very humid conditions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 35

The air quality anomaly is related to particulate matter (PM2.5) with a concentration of 4.7 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 35

Near 100 percent cloud cover suppressed photolysis rates, favoring the accumulation of NO_2 while ozone concentrations remained suppressed.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 35

Surface wind speed dropping to calm conditions alongside reduced planetary boundary layer heights limited atmospheric ventilation and trapped local pollution.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 35

The tceq NO_2 monitor at coordinates (29.583, -95.016) recorded 11.2 ppb at 2026-06-15T23:00:00+00:00, compared with an expected value of 0.6963855421686747 ppb, corresponding to a z-score of 17.497559719091377 and severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 35

At 2026-06-15T23:00:00+00:00 near the anomaly location, openaq ozone was 0.012 ppm, openaq PM2.5 was 2.4 ug/m3, purpleair PM2.5 was 4.7 ug/m3, and nearby tceq CO was 0.2 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 35

TCEQ monitoring detected an NO2 anomaly of 11.2 ppb at coordinates (29.583, -95.016) at 23:00 UTC on June 15, 2026, substantially higher than the baseline expected value of 0.70 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 35

Surface wind speeds decelerated to 0.0 m/s while planetary boundary layer height dropped from over 400 meters at 18:00 UTC to 152 meters around 00:00 UTC on June 16.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 35

On June 15, 2026, at 23:00 UTC, TCEQ monitoring station 482011050 located at (29.583, -95.016) recorded a severe nitrogen dioxide concentration of 11.2 ppb against an expected value of approximately 0.70 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 35

TCEQ monitor data showed a sharp NO2 spike to 11.2 ppb while SO2 concentrations remained near zero, suggesting a localized nitrogen oxides combustion source.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 35

A broad combustion plume or sulfur-rich industrial event is contradicted because concurrent co-pollutants did not rise similarly, with TCEQ CO near 0.2 ppm, TCEQ SO2 near -0.1 ppb, and no contemporaneous satellite column enhancement available at the anomaly time from Sentinel-5P.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 35

Regional wildfire smoke, dust, or a sulfur-rich industrial plume is a less plausible cause of the NO2 anomaly because ozone was low at the time of the spike, PM2.5 and PM10 were not elevated near the event, and available sulfur indicators did not show a concurrent increase.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 35

A localized primary NOx emission source near the anomalous TCEQ monitor is supported because NO2 at the site reached 11.2 ppb against an expected 0.70 ppb at 2026-06-15 23:00 UTC while nearby ozone was only 0.012 ppm and nearby PM2.5 was only 2.4 to 4.7 ug/m3, which is consistent with fresh NO titration and a source confined more to gases than to particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 35

The anomaly occurred on June 15th, 2026, at approximately 23:01 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 35

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to an accumulation of pollutants in the area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 35

The tceq NO2 network 6-hour mean increased from 7.64 ppb during 2026-06-15 18Z to 10.77 ppb during 2026-06-16 00Z, placing the anomaly within a broader evening increase in NO2 in the surrounding area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 35

NO2 levels were also elevated, with a mean value of 11.2 ppb, suggesting that there was an increase in nitrogen dioxide emissions from nearby sources.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.